

Turbidity Immersion Sensor

90S631100 · 90S631130

90S731100 · 90S731130



The digital immersion sensor is used for optical turbidity measurement according to the 90 ° IR scattered light method in raw-, waste- and process waters up to 4000 NTU.

Benefits

- Reliable concentration measurements by optical methods
- Pulsed infrared scattered light procedure
- No mechanically moving parts
- Digital reading
- Preprocessing in the sensor increases measurement sensitivity

Applications

- Measurement of turbidity in drinking water, domestic water, circulating water

Accessories

- Cable: Extension cables of 0.3 m, 2 m, 10 m, 25 m
- Controller: TriBox3, TriBox Mini, HS100

Technical Specifications

OPERATION AND SYSTEM CONFIGURATION

Measurement principle	90 ° scattered light method
Measuring method	Nephelometry

AUXILIARY POWER

Electrical connection	8-pin M12 plug
Power supply	12...24 V
Power consumption	3 W

INPUT PARAMETERS

Measuring ranges	0...40, 0...400, 0...1000, (0...4000 opt.) NTU
Cable specification	black PUR (halogen free), shielded, M12 plug
Measurement wavelength	880 nm

OUTPUT SIZES

Accuracy	± 2 % FS
Data interface	RS-485, Modbus RTU

PERFORMANCE CHARACTERISTICS

Response time	90 % of the value in 5 s
Accuracy	98 %
Calibration method	On controller, through analytical multipoint determination

AMBIENT CONDITIONS

Protection type	IP68
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PROCESS CONDITIONS

Process temperature	0...+60 °C
Process pressure	4 bar

STRUCTURAL DESIGN

Dimensions (Ø x L)	42 mm x 207 mm
Materials	Black PVC or stainless steel body, special glass optics, Viton® O-rings
Thread	1" GAS